

Comparison of Variable Selection and Regularization Methods

1. Introduction

Biomedical regression often involves numerous correlated predictors, complicating model selection and interpretation. Classical methods like stepwise regression are simple but unstable with multicollinearity or high dimensionality.

Ridge regression and LASSO address these challenges through regularization. Ridge applies an L2 penalty that shrinks coefficients while retaining all predictors, whereas LASSO uses an L1 penalty that forces some coefficients to exactly zero, enabling automatic variable selection. Their relative effectiveness depends on data structure, including dimensionality, correlation patterns, and signal sparsity.

This study compares Stepwise regression, Ridge regression, and LASSO across three cancer-related datasets representing low-, moderate-, and high-dimensional settings: the Prostate Cancer clinical dataset, the Breast Cancer Wisconsin dataset, and the METABRIC gene expression dataset. The analysis focuses on three central research questions: (1) under which conditions each method achieves superior predictive performance, variable selection accuracy, or model stability; (2) how the bias–variance tradeoff differs between Ridge and LASSO; and (3) in what scenarios Stepwise regression remains competitive despite its limitations. By integrating empirical results across datasets with distinct correlation structures and sparsity patterns, this work provides practical guidance for selecting appropriate regression methods in biomedical and applied statistical research.

2. Datasets and Analytical Strategy

2.1 Methodology Overview

In high dimensional biomedical data, predictor sets often exhibit strong multicollinearity and may contain comparable or even far more variables than the sample size. Hence, the analysis applies three approaches separately on three different datasets we chose. These methods operate under the same standard linear regression model:

$$y = X * \beta + \epsilon, \quad \epsilon \sim N(0, \sigma^2 I)$$

And they differ in how they constrain or penalize the coefficient β .

(1) Stepwise Regression. It attempts to identify a sparse subset of predictors by iteratively adding or removing variables according to an information criterion

such as AIC or BIC. Whatever specific penalty we choose, the goal is basically interpreted as:

$$\|y - X\beta\|^2 + \lambda\|\beta\|_0, \quad \|\beta\|_0 = \#\{j: \beta_j \neq 0\}$$

And j is constrained by the criterion we chose.(like AIC)

(2) Ridge Regression. It stabilizes the coefficient estimates by adding an L2 penalty. The modulus of the coefficient vector times λ , the optimization goal is:

$$\hat{\beta}_{Ridge} = \arg \min_{\beta} \|y - X\beta\|^2 + \lambda\|\beta\|_2^2$$

The original constraint for the coefficient β is $\|\beta\|_2^2 \leq t^2$, then the optimization process can be seen as letting the elliptical loss contour expand enough so as to intersect the boundary of the L2 ball. Therefore, the Ridge method tends to behave: **Shrink** coefficients to 0; **No coefficients** will be constrained to 0; **Reduce variance** when the coefficients are **highly correlated**.

(3) Lasso Regression.

$$\hat{\beta}_{Lasso} = \arg \min_{\beta} \|y - X\beta\|^2 + \lambda\|\beta\|_1$$

Similar as Ridge, Lasso adds an L1 penalty. And the original constraint for the coefficient β is $\|\beta\|_1 \leq t$, then the optimization process can be geometrically interpreted as letting the elliptical contour expand to touch the feasible region which is a cross-polytope. It is likely that it will intersect the feasible region first on a sharp corner on coordinate axes. Therefore, the Lasso method tends to behave: **Automatically** choose the variable; **Model size** controlled by t ; **High interpretability** in **high dimensional** settings.

2.2 We analyzed three distinct datasets:

(1)Prostate Cancer Clinical Data (n=97, p=8): 97 patients, 8 predictors. Low-dimensional but with highly correlated clinical measures.

(2)Breast Cancer Dataset(n=569, p=30): clinical variables, used to build Ridge Logistic and test Stepwise stability.

(3)Breast Cancer Gene Expression Profiles (METABRIC)(n≈1900, p=693): ~1900 patients, ~500 gene expression predictors. High dimensional gene expression data predicting a continuous prognostic index (Nottingham Prognostic Index). Used for Ridge vs. LASSO comparison.

We will first discuss our findings with Stepwise regression, then present a detailed comparison of Ridge and LASSO on the genomic data.

2.3 Data preparation

- **Prostate cancer dataset:** The original file contains an indicator column specifying the recommended training subset. We split the data accordingly and remove the train column so that it is not used as a predictor. The response is the log PSA level (lpsa), and the eight clinical covariates serve as predictors.

Also, this project computes Cook's distances for full linear model. The largest Cook's distance was 0.21, which is above the $\frac{4}{n}$ threshold but much below 1. This indicates a few moderately influential observations, but no extremely dominating outliers, so we kept all the points in the main analysis.

- **Breast cancer dataset:** It includes one ID column, a diagnosis label(B/M), and 30 numeric predictors. This project removes the ID column and transfers the label to a binary indicator, and randomly pick 60% data as train dataset and 30% data as test dataset. Before applying ridge regression, this project generates model matrices to standardize 30 numeric predictors.

For this dataset, this project focuses more on train-test splitting and cross validation to evaluate the model performance rather than outliers or influential points. Ridge regression is more robust to outliers comparing to stepwise selection because of L2 shrinkage.

3. Stepwise Regression

3.1 Initial Model (Dataset 1)

We began the prostate analysis by fitting a full linear model with all 8 predictors. The key results are as follow: The model is significant (F-test $p \approx 2.0e-12$, $R^2=0.69$). Among individual covariates, lcavol and lweight are strongly significant, and lbph and svi are also significant at the 5% level, whereas age, lcp, gleason, and pgg45 are not statistically significant in this joint model.

This suggests some predictors may be redundant, motivating the use of a selection method like Stepwise to find a simpler model. (Same as Lasso)

3.2 Stepwise Regression Results

AIC Criteria

Next, we run stepwise regression using AIC. Here's what happens: the model ends up keeping 7 predictors – it only drops gleason. The AIC score improves from -37.13 to -39.10, so the fit gets better.

step_aic\$anova					
Step <chr>	Df <dbl>	Deviance <dbl>	Resid. Df <dbl>	Resid. Dev <dbl>	AIC <dbl>
	NA	NA	58	29.42638	-37.12766
- gleason	1	0.01091586	59	29.43730	-39.10281

Figure 1

BIC Criteria

Next, we switch to BIC, which is stricter about adding variables. The final model keeps only 2 predictors: leavol and lweight. The model explains about 61% of the variation in log-PSA. The fit also gets better as the score improvement shows.

Step <chr>	Df <dbl>	Deviance <dbl>	Resid. Df <dbl>	Resid. Dev <dbl>	AIC <dbl>
	NA	NA	58	29.42638	-17.28543
- gleason	1	0.01091586	59	29.43730	-21.46527
- age	1	1.10247781	60	30.53978	-23.20654
- lcp	1	1.52966920	61	32.06945	-24.13670
- pgg45	1	0.74554742	62	32.81499	-26.80161
- lbph	1	2.09275411	63	34.90775	-26.86414
- svi	1	2.18409678	64	37.09185	-27.00272

Figure 2

So here's the issue, the analysis used the same data and the same method, but the different criteria(AIC vs. BIC) gave us totally different models: 7 variables vs. 2variables. This suggests that stepwise regression depends on the specific criteria too much.

4 Regularization Methods

4.1 Stepwise Regression on Breast Cancer Data

While Stepwise Regression produced simplified models, we tested its stability using bootstrap resampling. The results from the Breast Cancer Data shows that after running stepwise on many resampled datasets, no single variable was consistently selected (the selection frequency of all 30 predictors can't exceed 80%).

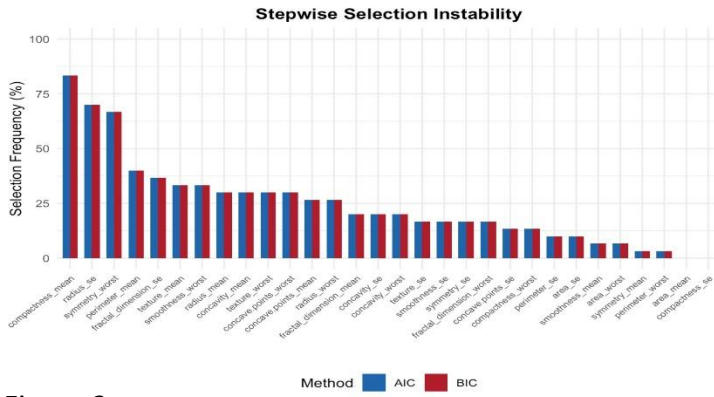


Figure 3

random sample variation. For Dataset 2, the set of "selected" predictors changed each time we resampled. Variables would come in and out of the model almost randomly, there was no consistent signal. Stepwise is unstable under certain circumstances like biomedical data.

Figure below displays the correlation heatmap of 30 numeric predictors in the training dataset. The matrix reveals clear blocks of strong positive correlation such as radius and area (correlations are above 0.8). Also, different versions of features (se and worst) are strongly correlated. This phenomenon indicates substantial multicollinearity among the predictors, suggesting the instability of stepwise selection, which motivates the ridge regression which is used to handle correlated covariates.

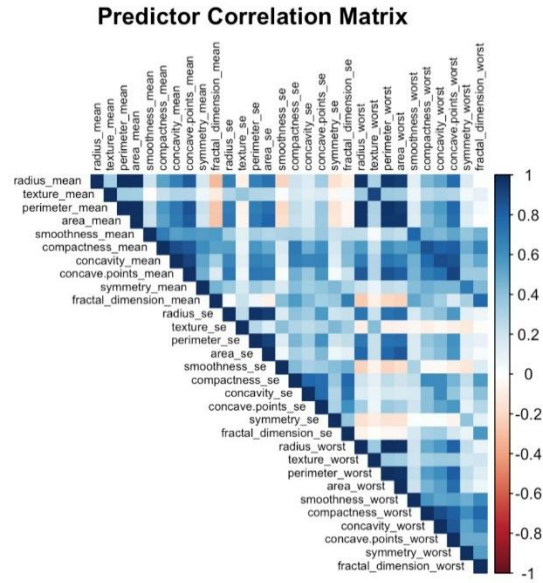


Figure 4

4.2 Ridge Logistic Regression Model

For the training data we fit a logistic regression model with an ℓ_2 penalty using "glmnet". The estimator minimizes the penalized negative log-likelihood:

$$\frac{1}{n_{train}} \sum_i \{-y_i \log \pi_i - (1 - y_i) \log(1 - \pi_i)\} + \lambda \sum_{j=1}^p \beta_j^2,$$

Where $\pi_i = \Pr(Y_i = 1|X)$ and λ controls the shrinkage. This report uses 10-fold cross-validation to select λ . CV curve shows that Deviance increases slowly with λ , showing Ridge effectively stabilizes estimates; Best λ (min) is 0.0386 and 1SE λ is 0.0614(Appendix A). All 30 numeric predictors maintain nonzero coefficients but they all shrink because of ℓ_2 penalty. As λ increases, the predictors highly related to ‘size’ and ‘shape’ have lower coefficient values, which contributes to resolve the multicollinearity observed in the correlation matrix.

To evaluate predictive performance, this report converts the predicted probabilities to a 0.5 threshold. The accuracy for test dataset improves to 96.9%, indicating that ℓ_2 regularization is beneficial to generalization without being sensitive to the exact tuning parameter value.

Overall, in Breast Cancer Dataset, predictors are highly correlated, making traditional logistic regression or Stepwise methods unstable. Ridge logistic regression with ℓ_2 penalty effectively handles multicollinearity, retains all variables, and delivers stable classification performance. This further supports the use of regularization methods with correlated predictors in clinical datasets.

4.3 Regularization: METABRIC Dataset

The METABRIC dataset presents a particularly challenging modeling environment due to its very high dimensionality and the presence of noisy, redundant, and highly correlated gene expression predictors. In such contexts, it is natural to question whether Ridge regression can still perform effectively, or whether LASSO, with its L1 penalty and inherent variable-selection capability, offers a superior alternative. To address this question, both Ridge and LASSO regression were applied to the METABRIC dataset, and their predictive accuracy and bias–variance characteristics were compared.

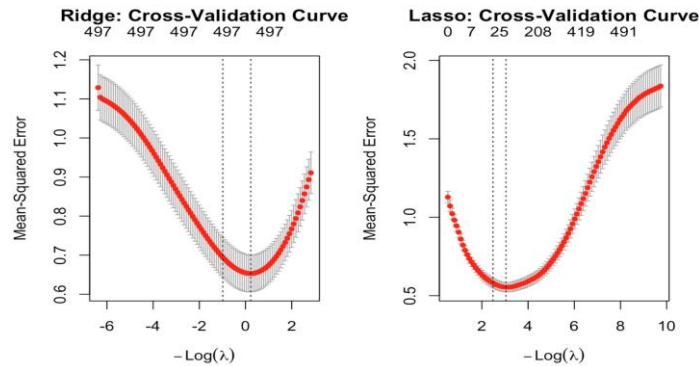


Figure 5: Ten-fold cross-validation curves for Ridge (left) and LASSO (right) regression models on the METABRIC dataset. The red curve represents the cross-validated mean squared error (MSE) across a range of penalty values (λ), with the gray bands indicating variability across folds. The vertical dotted lines mark the λ values corresponding to the minimum MSE and the 1-SE rule.

Nottingham Prognostic Index (NPI) served as the continuous outcome variable, and ten-fold cross-validation was used to select the optimal tuning parameter for each method. Cross-validated mean squared error (MSE) results (Figure 5) showed a clear performance difference: LASSO achieved a lower MSE of approximately 0.50, while Ridge regression produced a higher MSE of approximately 0.61. This result indicates that LASSO provides better predictive performance for this high-dimensional genomic dataset.

The contrasting performance of the two methods reflects their different approaches to managing model complexity. LASSO, through its L1 penalty, shrinks a large number of coefficients exactly to zero and ultimately retained only 47 predictors. Ridge, by contrast, applies an L2 penalty that shrinks coefficients continuously but retains all predictors; in this analysis, all 497 predictors remained in the model (Figure 6). The coefficient paths (Figure 7) further illustrate this distinction: Ridge coefficients shrink smoothly toward zero, whereas LASSO coefficients collapse to zero abruptly, producing a sparse and more interpretable model. Such sparsity is particularly important in genomic settings, where only a small subset of genes is expected to have biological or prognostic relevance.

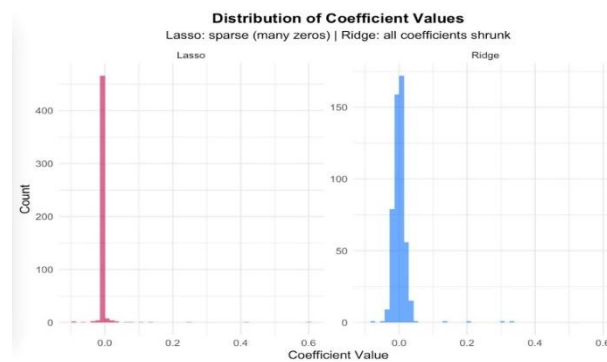


Figure 6: Distribution of coefficient estimates for LASSO (left) and Ridge (right) regression fitted to the METABRIC dataset.

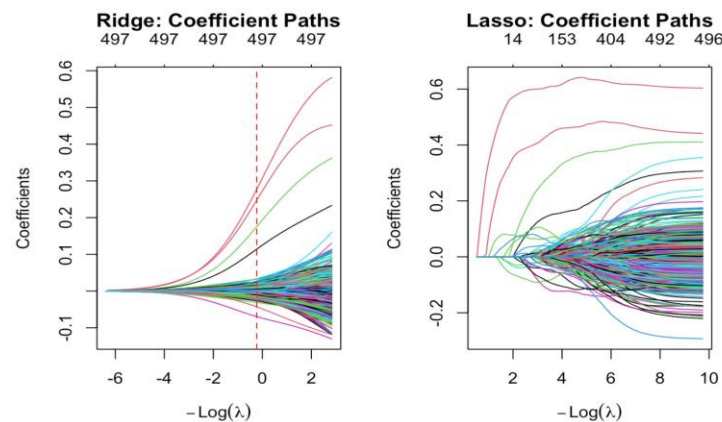


Figure 7: Coefficient paths for Ridge (left) and Lasso (right) regression as a function of the regularization parameter λ . The vertical dashed line indicates the optimal λ selected by cross-validation. Numbers at the top show the count of non-zero coefficients at different λ values.

The bias–variance tradeoff (Figure 8) further clarifies these findings. Ridge regression exhibited lower training error, consistent with lower bias. However, a substantial gap between training and test error suggested high variance and overfitting. LASSO displayed slightly higher training error but nearly identical training and test error, indicating substantially lower variance. Consequently, the total test error for LASSO was lower than that of Ridge, confirming better generalization performance.

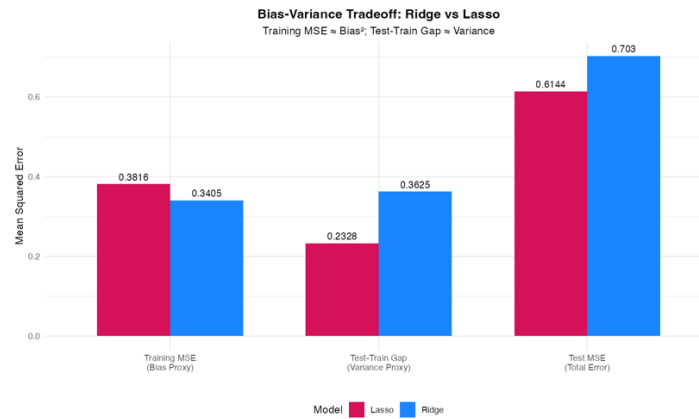


Figure 8: Bias-variance decomposition for Ridge and Lasso regression models.

Standard predictive performance metrics (Figure 9) also favored LASSO. On the held-out test data, LASSO achieved a lower RMSE (0.71) compared with Ridge (0.79), and a higher test R^2 value (0.55 vs. 0.45). These results demonstrate that although Ridge fits the training data more closely, it predicts future observations less accurately than LASSO. Furthermore, scatterplots of predicted versus actual NPI values (Figure 10) show that LASSO predictions cluster tightly around the identity line, while Ridge predictions exhibit greater scatter, especially at higher and lower NPI values. This suggests that, in clinical settings where prognostic accuracy is essential, LASSO would likely yield more reliable predictions for new patients.

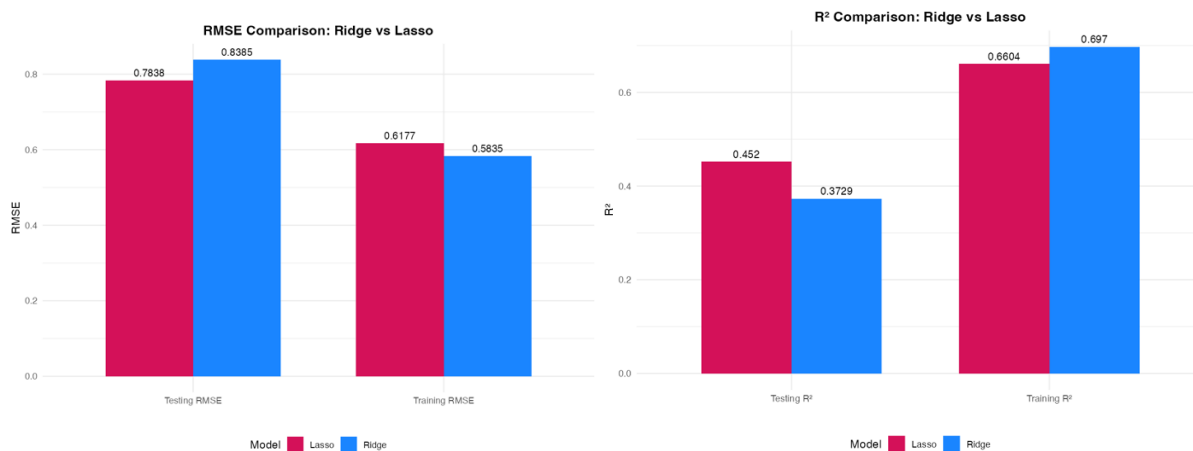


Figure 9: RMSE (left) and R^2 (right) comparison between Ridge and Lasso regression. Lasso demonstrates superior predictive performance with lower test RMSE (0.71 vs 0.79) and higher test R^2 (0.55 vs 0.45).

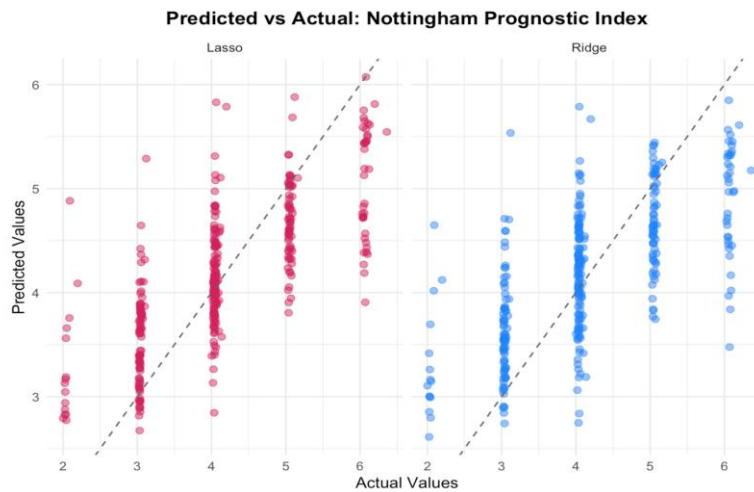


Figure 10: Predicted vs. actual Nottingham Prognostic Index (NPI) values for Lasso (left) and Ridge (right) regression. Lasso predictions cluster more tightly around the identity line, indicating better predictive accuracy, while Ridge shows greater dispersion especially at extreme values.

Overall, the METABRIC analysis demonstrates that while Ridge regression stabilizes coefficients under multicollinearity, it retains too many noisy predictors in high-dimensional contexts, resulting in poorer predictive performance. LASSO, by contrast, achieves effective variable selection, produces a sparse and interpretable model, and demonstrates superior generalization. For this dataset, LASSO was the most appropriate regularization method, offering the best combination of accuracy, stability, and interpretability.

Discussion

This study addressed three central research questions regarding the comparative performance of Stepwise regression, Ridge regression, and LASSO across low-, moderate-, and high-dimensional biomedical datasets.

Predictive Performance, Variable Selection, and Model Stability

Clear patterns emerged across the three datasets. Stepwise regression produced simple, interpretable models in the low-dimensional Prostate Cancer dataset but demonstrated substantial instability when predictors were correlated or when sample size was limited. Small changes in data or resampling yielded drastically different selected models in both the Prostate and Breast Cancer datasets. Ridge regression effectively stabilized estimates through coefficient shrinkage, providing reliable results in the presence of multicollinearity. However, its retention of all predictors became a limitation in the high-dimensional

METABRIC dataset, where many predictors were noisy or irrelevant. LASSO proved most effective in the high-dimensional setting, removing irrelevant predictors while preserving informative ones, achieving the lowest cross-validated error and superior prediction accuracy.

Bias–Variance Tradeoff

Ridge and LASSO exhibited distinct bias–variance characteristics. Ridge regression reduced variance by shrinking correlated coefficients but retained all predictors, leading to higher variance in high-dimensional settings like METABRIC. LASSO's L1 penalty introduced slightly more bias through aggressive coefficient shrinkage but substantially reduced variance by enforcing sparsity, resulting in better generalizability when the underlying signal was sparse.

Competitiveness of Stepwise Regression

Despite its limitations, Stepwise regression remained competitive in the low-dimensional Prostate Cancer dataset where predictor correlations were modest. However, its instability and sensitivity to data perturbations make it unreliable for modern biomedical contexts involving correlated or high-dimensional predictors.

Conclusion

Overall, the comparative analysis shows that model selection and regularization methods must be chosen based on the dimensionality, correlation structure, and underlying signal complexity of the dataset. Stepwise regression remains useful for small, low-correlation datasets but is unreliable in modern biomedical contexts. Ridge regression is effective when all predictors may contribute to the outcome and multicollinearity is a concern, but its tendency to retain all variables limits its utility in high-dimensional settings. LASSO emerges as the preferred method for genomic or other high-dimensional data, offering strong predictive performance, reduced overfitting, and meaningful feature selection. These findings provide practical guidance for researchers selecting statistical tools for clinical and biomedical applications.

Appendix

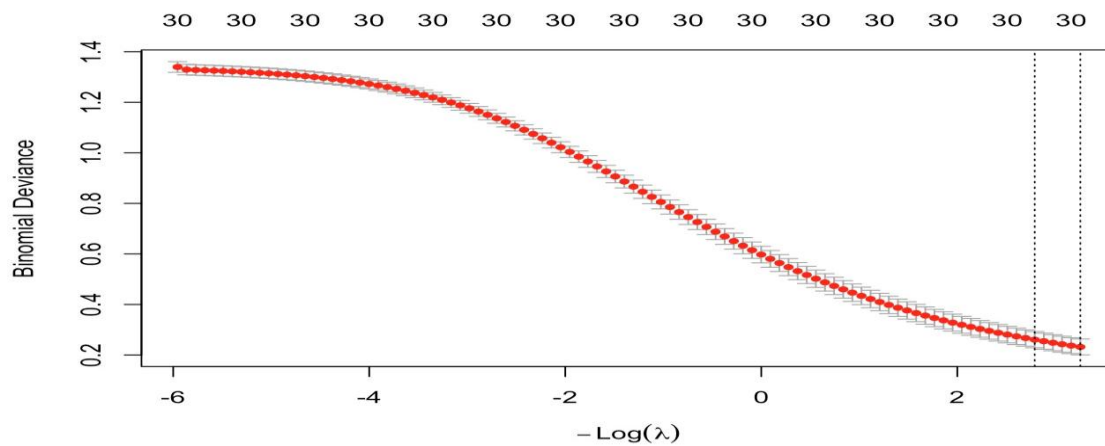
A. CV curve for Ridge regression

```
ridge$lambda.min
```

```
## [1] 0.03857954
```

```
ridge$lambda.1se
```

```
## [1] 0.06142954
```



B. The code of this report:

<https://github.com/RuichenTang/STAT-527-Final-project.git>